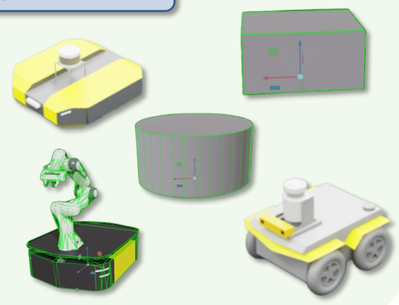
# Dataset « RobotPath-Bench » Evaluation Protocol

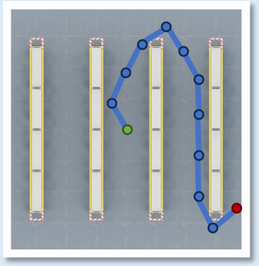
5 Robots



7 Layouts



Path



13,924 Collisions

20,113 Eligible Records

20,399 Path-Execution Records

Simple Collection
4,745 Collisions

900 items

653 Collisions

Hard Collection
9,179 Collisions

1,200 items

740 Collisions

Simple Collection 7K+



Turn Translate

Hard Collection 13K+



Turn + Translate

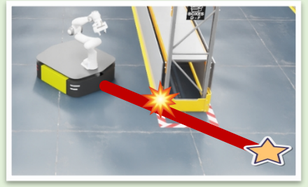
Layout Transfer
Robot: Seen
Layout: Unseen

Robot Transfer
Robot: Unseen
Layout: Seen

Joint Transfer
Robot: Unseen
Layout: Unseen

Configuration-Transfer

Task 1: Pre-Collision Path Diagnosis

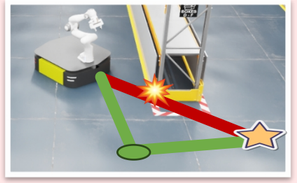


Classification
Balanced Accuracy & F1

Identification
Object Accuracy
Segment Accuracy

Localization
Contact Hit@0.1
Contact Hit@0.2
Mean Contact Error

Task 2: Local Repair



Constructibility
Context Coverage

Success
Top-1, Best-of-8
Fallback Attempt Rate
Final Repair Success

Quality
Target Retention
Target Displacement
Clearance, Number of Points, Path Increase